

Project Design Phase-II

Data Flow Diagram & User Stories

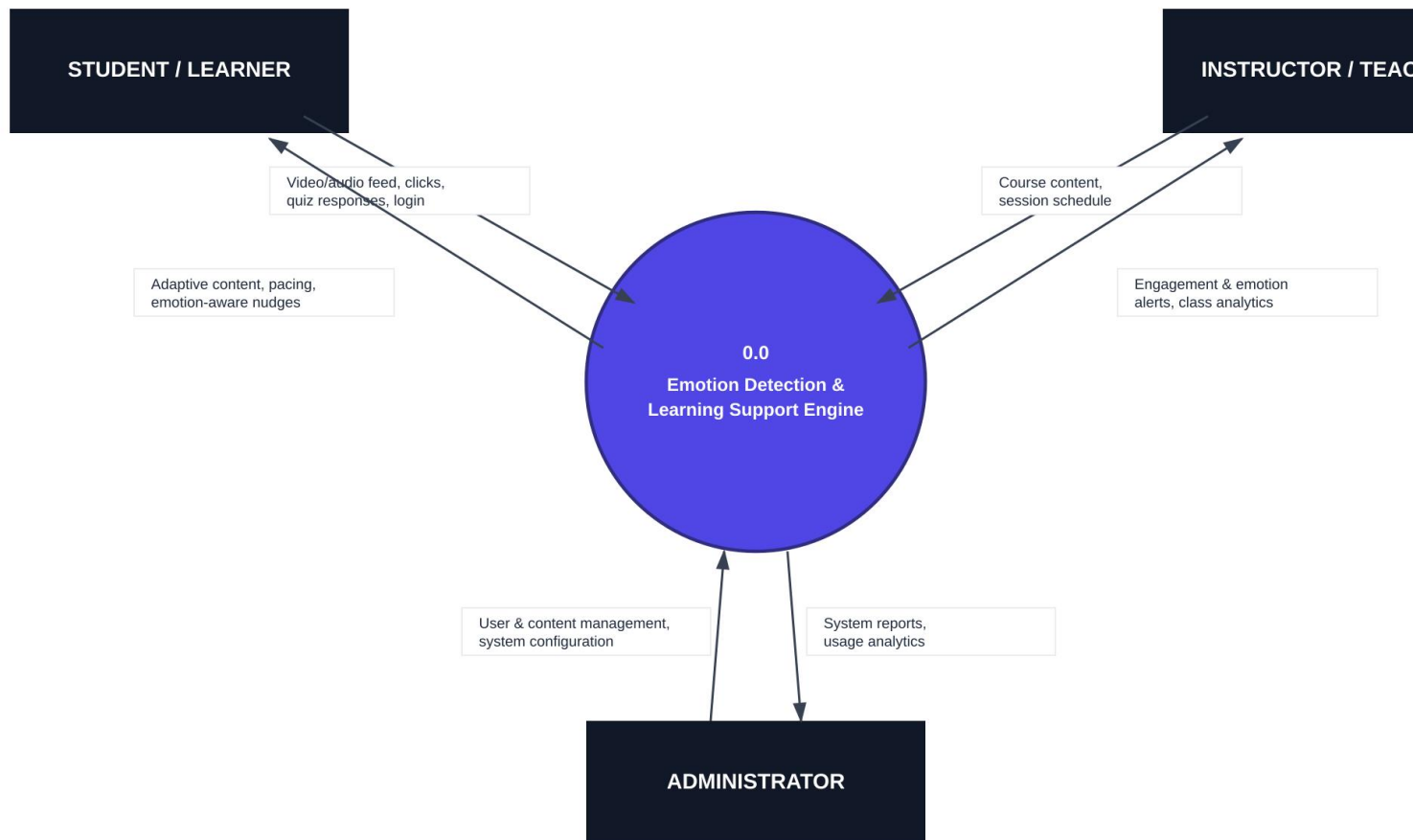
Date	07 July 2026
Team ID	SWTID-2026-2247
Project Name	Emotion Detection and Learning Support Engine
Maximum Marks	4 Marks

Data Flow Diagrams:

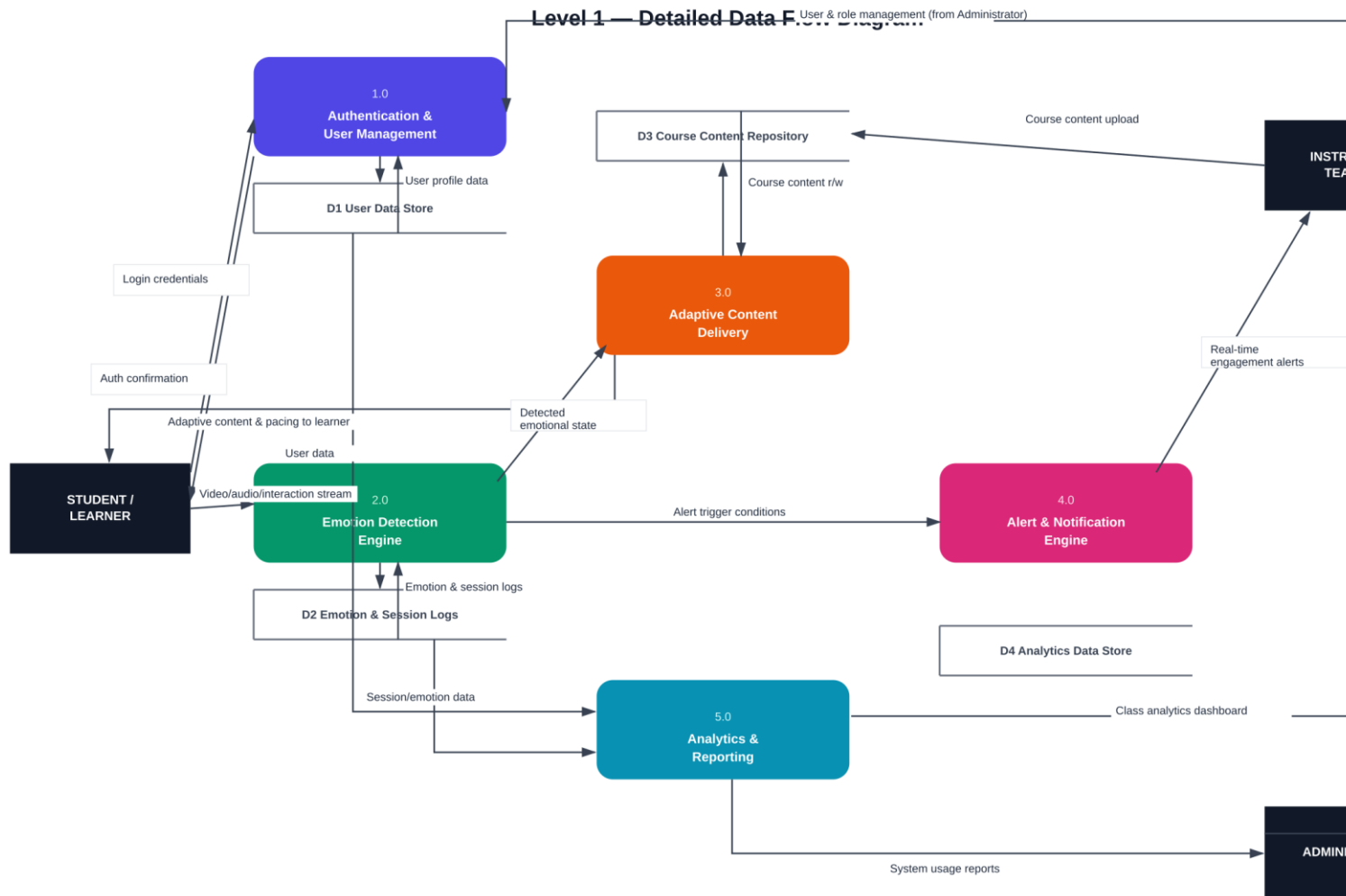
A Data Flow Diagram (DFD) shows how information moves through the Emotion Detection and Learning Support Engine — how data enters and leaves the system, what transforms it, and where it is stored. Below are the Level 0 (context) and Level 1 (detailed) diagrams for the system.

Level 0 — Context Diagram

Level 0 — Context Diagram



Level 1 — Detailed Data Flow Diagram



User Stories

User stories for the Emotion Detection and Learning Support Engine, covering the Student (Learner), Instructor (Teacher), Administrator, and the System (AI Engine) itself.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Student (Learner)	Registration	USN-1	As a learner, I can register by entering my name, email, password, and confirming my password.	I can create an account and land on my personalized dashboard.	High	Sprint-1
		USN-2	As a learner, I receive a confirmation email once I register.	I receive an email and can confirm my account via a verification link.	High	Sprint-1
		USN-3	As a learner, I can register/log in using Google or institutional SSO.	I can access my dashboard using Google/SSO login.	Medium	Sprint-2
	Login & Consent	USN-4	As a learner, I can log in with my email and password.	I am authenticated and redirected to my dashboard.	High	Sprint-1
		USN-5	As a learner, I am asked for explicit consent before my camera/microphone is used for emotion detection.	I see a clear consent prompt and can opt in/out before any capture begins.	High	Sprint-1
	Emotion Capture	USN-6	As a learner, the system observes my facial expressions and engagement during a lesson to detect confusion, boredom, or stress.	My emotional/attention state is detected in real time with at least 80% accuracy in testing.	High	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
		USN-7	As a learner, I can pause or disable emotion tracking at any time during a session.	A visible toggle lets me turn tracking off instantly without losing my session.	Medium	Sprint-2
	Adaptive Learning	USN-8	As a learner, I receive simplified content or a short recap automatically when I appear confused.	Simplified content/recap appears within seconds of confusion being detected.	High	Sprint-3
		USN-9	As a learner, I get a short break suggestion or an engaging activity when I appear bored or fatigued.	A break/engagement prompt is shown after sustained low-engagement is detected.	Medium	Sprint-3
		USN-10	As a learner, I can view my own emotional/engagement trends over time.	I can see a simple chart of my focus and mood trends across past sessions.	Low	Sprint-4
	Dashboard	USN-11	As a learner, I can view my course progress, pending tasks, and upcoming deadlines on one dashboard.	Dashboard loads with progress %, pending tasks, and deadlines correctly listed.	Medium	Sprint-2
Instructor (Teacher)	Course Management	USN-12	As an instructor, I can upload and organize course content (videos, slides, quizzes).	Uploaded content appears correctly under the right module/course.	High	Sprint-1
	Live Alerts	USN-13	As an instructor, I receive a real-time alert when multiple students appear confused or disengaged during a live session.	An alert appears on my dashboard within seconds of the threshold being crossed.	High	Sprint-3
		USN-14	As an instructor, I can see which specific topic triggered a confusion spike.	The alert links directly to the timestamp/slide that caused the spike.	Medium	Sprint-3
	Class Analytics	USN-15	As an instructor, I can view a class-level analytics dashboard showing aggregate engagement and emotion trends.	Dashboard shows engagement graphs per session, updated after each class.	High	Sprint-4
		USN-16	As an instructor, I can export session analytics as a report (PDF/Excel).	I can download a correctly formatted report for any selected session.	Low	Sprint-4
Administrator	User Management	USN-17	As an administrator, I can create, edit, and deactivate learner and instructor accounts.	Account changes are saved and reflected immediately in the system.	High	Sprint-1
	System Configuration	USN-18	As an administrator, I can configure emotion-detection sensitivity thresholds and alert rules.	Changed thresholds take effect for all new sessions without a restart.	Medium	Sprint-3
	Reporting	USN-19	As an administrator, I can view platform-wide usage and performance reports.	Reports show accurate active-user counts, engagement averages, and system uptime.	Medium	Sprint-4
	Data Privacy	USN-20	As an administrator, I can review and manage data retention/deletion settings for emotion data.	I can set a retention period and trigger deletion in compliance with policy.	High	Sprint-2
System (AI Engine)	Model Monitoring	USN-21	As the system, I log model confidence scores for each emotion prediction for later review.	Every prediction is logged with a confidence score and timestamp.	Medium	Sprint-3
	Failover	USN-22	As the system, I fall back to a neutral/no-adaptation mode if the emotion-detection service is unavailable.	Learners continue their session uninterrupted even if detection fails.	High	Sprint-2